Food composition triggers early sex differentiation of a shrimp: a transcriptomic analysis

F Glaviano[1], E Somma[1], M Mutalipassi[1], N Ruocco[2], T Viel[1], P Nieri[3], F Esposito[2], ED Aflalo[4], A Sagi[4], V Zupo[1], M Costantini[2]

[1]Stazione Zoologica Anton Dohrn, Department of Marine Biotechnology, Villa Dohrn, Ischia (Italy); [2]Stazione Zoologica Anton Dohrn, Department of Marine Biotechnology, Villa Comunale, Napoli (Italy); [3]Department of Pharmacy, Pisa University (Italy); [4]Department of Life Sciences, Ben-Gurion University of the Negev, Israel.

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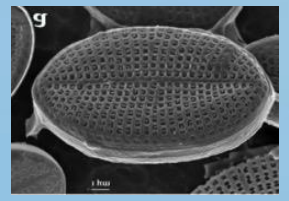
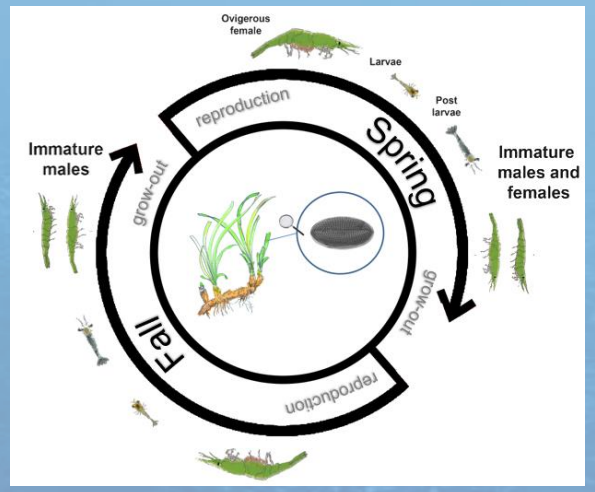
The shrimp *Hippolyte inermis* undergoes a well-known process of sex reversal triggered by the ingestion of benthic diatoms. The structure of the wound-activated diatom's compound is still under elucidation, but research is ongoing to clarify its very selective mechanism of action.

This shrimp co-evolved with diatoms *Cocconeis* spp. seasonally present on the leaf stratum of various seagrasses and in particular in *Posidonia oceanica* meadows [1].



The peculiar reproductive strategy of the shrimp, seasonally alternating two different sexual determination phases, protandric in fall, and apparently gonochoristic in spring, facilitates the survival in a highly competitive benthic environment [1-2].



The mechanisms controlling sexual differentiation in crustacean have not been fully revealed, but a key role has been attributed to the insulin-like hormone (IAG) produced by the androgenic gland (AG), a crustacean masculine endocrine organ [3].

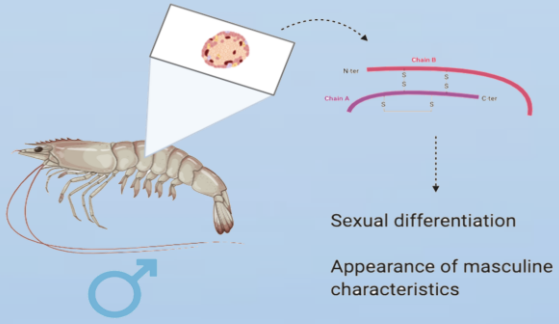
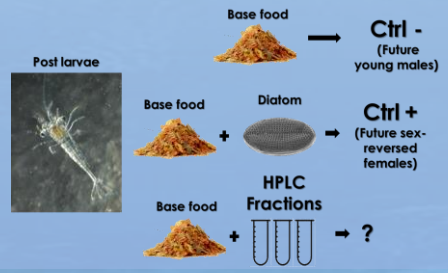


Figure from Sharabi et al [3]

Physiologically, it has been suggested that compounds present in *Cocconeis* spp. diatoms can be responsible of apoptosis of the androgenic gland (AG) and, consequently, control the sexual shift from male to female [4].

The chemical structure of this cell death compound has not yet been identified but is under investigation and, since this shrimp is so far the only biological "sensor" able to track the presence of this active compound, the traditional bioassay consists of feeding young post larvae with different fractions obtained through HPLC separations.



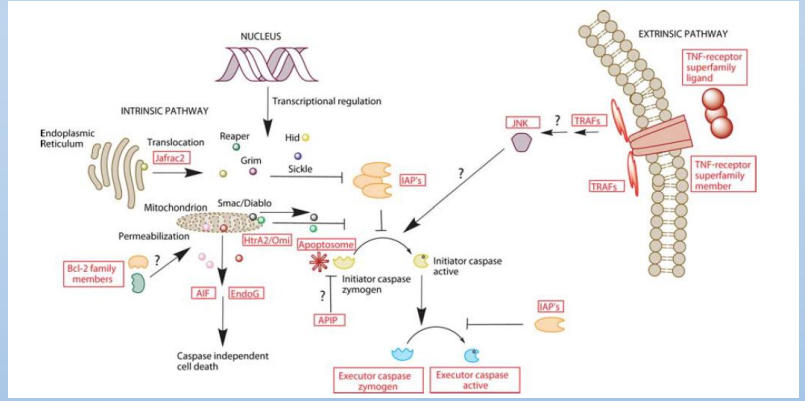


Figure from Menze et al [5]

A *de novo* transcriptome analysis performed in the frame of this research reveals that most genes affected by the experimental administration of diatoms to young post-larvae are involved in the apoptosis pathway and IAG secretion pathway (paper in preparation). This is in line with the universal mechanism of control of crustacean sexual differentiation by the insulin-like androgenic hormone secreted by the AG. However, new details emerged from this study will aid the comprehension of the actual mechanism of action of the diatom's compound, to be validated through real time experiments, still in progress.

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